

FBE 430 - Introduction to Decentralized Finance

Sample exam

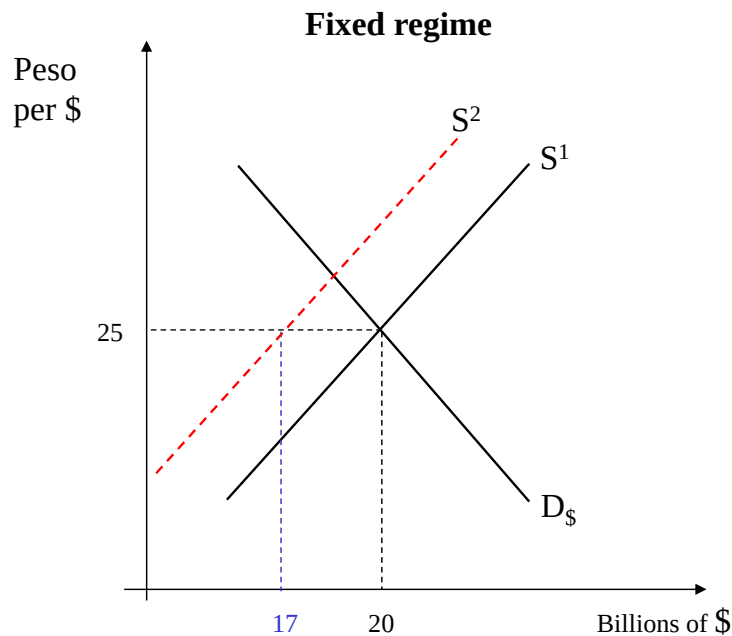
Wednesday May 27, 3:00-3:50PM

1. DeFi solves some of the problems associated with traditional finance by
 - (a) Reducing the automatization of the functions played by the middlemen or intermediary.
 - (b) Increasing the centralized control of few players.
 - (c) Enhancing the accessibility of marginal costumers to financial services. **Right answer.**
 - (d) Reducing the interconnectivity of the financial system.

2. In a **public** blockchain network
 - (a) Access requires permission.
 - (b) Control over the network is governed by a single entity.
 - (c) There are no restrictions to who can access it. **Right answer.**
 - (d) Some information is freely available in the network while other requires permission.

3. Which of the following statements is correct?
 - (a) Utility tokens act as medium of exchange and as store of value.
 - (b) Payment tokens enable their holders to interact with or gain utility from a specific platform. Similar to tokens in an amusement park.
 - (c) Holders of payment tokens are entitled to certain cash flows such as dividends.
 - (d) Security tokens are linked to and constitute an economic claim to a real-world asset. **Right answer.**

4. What is the main advantage of stablecoins compared to other cryptocurrencies?
- (a) Transactions of stablecoins do not need to be recorded in a Blockchain. This save energy costs needed to run a blockchain.
 - (b) Stablecoins are covered by federal insurance and in the eventuality that their value crashes, investors will be reimbursed by the federal insurance. Regular cryptocurrencies, instead, are not insured.
 - (c) The prices of regular cryptocurrencies tend to be more volatile than the prices of stablecoins. **Right answer.**
 - (d) We can track the real identity of the owners of stablecoins but not of the owners of regular cryptocurrencies.
5. Suppose that Banco of Mexico is trying to keep the exchange rate Peso/Dollar at 25 (25 Pesos to buy one Dollar). The graph below plots the demand and supply of Dollars in exchange of Pesos. Suppose that the supply of dollars shifts from S_1 to S_2 . Which of the following statements is correct.



- (a) Without intervention from the Banco of Mexico the Peso would appreciate (we need less pesos to buy one dollar).
- (b) To keep the rate at 25 Banco of Mexico must purchase 3 billion Dollars.
- (c) To keep the rate at 25 Banco of Mexico increases the reserves of Dollars.
- (d) To keep the exchange rate at 25 Banco of Mexico must sell 3 billion Dollars. **Right answer.**

6. Consider two cryptocurrencies: A and B . The first cryptocurrency has an expected return of 6% and a standard deviation of return of 10%. The second cryptocurrency has an expected return of 10% and a standard deviation of return of 15%. The correlation of returns between the two securities is $\rho_{AB} = 0.4$. Suppose that you allocate 80% of your portfolio in cryptocurrency A and 20% in cryptocurrency B . What is the Sharpe ratio of this portfolio if the risk-free rate is 5%?

- (a) Sharpe ratio = 1.9522.
- (b) Sharpe ratio = 0.7082.
- (c) Sharpe ratio = 0.2590.
- (d) Sharpe ratio = 0.1874. **Right answer.**

The Sharpe ratio is given by $(r_p - r_f)/\sigma_p$ where r_p =expected return of the portfolio, r_f =risk-free rate, and σ_p is the standard deviation of the portfolio return. In the problem we know $r_f = 0.05$. So we need to compute r_p and σ_p . The expected return is $r_p = 0.8 \times 0.06 + 0.2 \times 0.1 = 0.068$. The standard deviation is

$$\sigma_p = \sqrt{0.8^2 \times 0.1^2 + 0.2^2 \times 0.15^2 + 2 \times 0.8 \times 0.2 \times 0.4 \times 0.1 \times 0.15} = 0.096028.$$

So we now have all we need to calculate the Sharpe ratio:

$$\text{Sharpe ratio} = \frac{0.068 - 0.05}{0.096028} = 0.1874$$

7. An investor owns 50 ETH, each worth 2,000 Dollars, in the MakerDAO protocol. The investor deposits 5 ETH in a Vault in order to mint DAI (each DAI worth 1 Dollar). The minimum collateralization ratio is 200%. What is the maximum number of DAI that the investor can mint?

- (a) 3,000 DAI
- (b) 5,000 DAI **Right answer.**
- (c) 7,000 DAI
- (d) 9,000 DAI

8. Suppose that the total supply in Compound is 600 million DAI. The total borrowed amount in Compound is 480 million DAI. The base rate is 0.2% ($a=0.002$) and the slope is 14% ($b=0.14$). What is the utilization rate?
- (a) 70%
 - (b) 80% **Right answer.**
 - (c) 65%
 - (d) 75%
9. Consider the DAI/USDC market in Uniswap with a supply of 1,025,000 DAI and 1,000,000 USDC. The exchange rate in Uniswap is 1.025 DAI for 1 USDC and the invariant is 1.025 trillions. On a different Decentralized Exchange (DEX), the price for USDC is 1.04 DAI. A trader could arbitrage with a flash swap. Which of the following statements is correct? **Remark:** For this question ignore slippage in the exchange.
- (a) The trader would like to buy USDC in the other DEX and sell them in Uniswap.
 - (b) If the trader starts the process by taking a loan in DAI with a flash swap in Uniswap, the trader will incur a loss for sure (rather than a profit)
 - (c) The trader purchases USDC in Uniswap and sell them in the other DEX. **Right answer.**
 - (d) If the trader starts the process by taking a loan in USDC with a flash swap in Uniswap, the trader will incur a loss for sure (rather than a profit).
10. In the MakerDAO protocol, an investor owns 50 ETH, each worth 2,000 Dollars. The investor deposits 25 ETH in a Vault in order to mint DAI (each DAI worth 1 Dollar). The minimum collateralization ratio is 200%. If the investor chooses to mint 20,000 DAI, what is the **actual** collateralization ratio?
- (a) 250% **Right answer.**
 - (b) 300%
 - (c) 400%
 - (d) 500%

11. In the MakerDAO protocol, an investor owns 50 ETH, each worth 2,000 Dollars. The investor deposits 10 ETH in a Vault in order to mint DAI (each DAI worth 1 Dollar). The minimum collateralization ratio is 200%. The investor mints 4,000 DAI. If the ETH price drops to 800 Dollars, how many ETH need to be added to the initial deposit of 10 ETH in order to have an **actual** collateralization ratio of 300%?
- (a) 1.00 ETH
 - (b) 5.00 ETH **Right answer.**
 - (c) 10.00 ETH
 - (d) 12.50 ETH
12. An investor deposits 5000 DAI and 10 ETH in Compound. DAI has a collateral factor of 80 and ETH has a collateral factor of 60. If 1 DAI = \$1 and 1 ETH=\$2,000, what is the total dollar value of the deposit (from both DAI and ETH deposits)?
- (a) \$10,000
 - (b) \$20,000
 - (c) \$25,000 **Right answer.**
 - (d) \$30,000
13. The total supply in Compound is 600 million DAI. The total borrowed amount in Compound is 450 million DAI. The base rate is 0.75% ($a=0.0075$) and the slope is 10% ($b=0.1$). What is the borrow rate?
- (a) 0.0725 or 7.25%
 - (b) 0.0825 or 8.25% **Right answer.**
 - (c) 0.0925 or 9.25%
 - (d) 0.0625 or 6.25%
14. The total supply in Compound is 400 million DAI. The total borrowed amount in Compound is 380 million DAI. The base rate is 0.8% ($a=0.008$). There is a kink in the slope at 0.85 (85%) utilization rate. The slope for utilization rates smaller than 85% is 10% ($b=0.1$). The slope for utilization rates bigger than 85% is 40% ($b=0.4$). What is the supply rate?
- (a) 0.126 or 12.6% **Right answer.**
 - (b) 0.133 or 13.3%
 - (c) 0.113 or 11.3%
 - (d) 0.143 or 14.3%